

Answer ALL questions.

1. Figure 1 shows a site map of an area in Guyana that is being developed as an industrial estate. The adjoining river was a good source of fish for local residents of the Community at B.

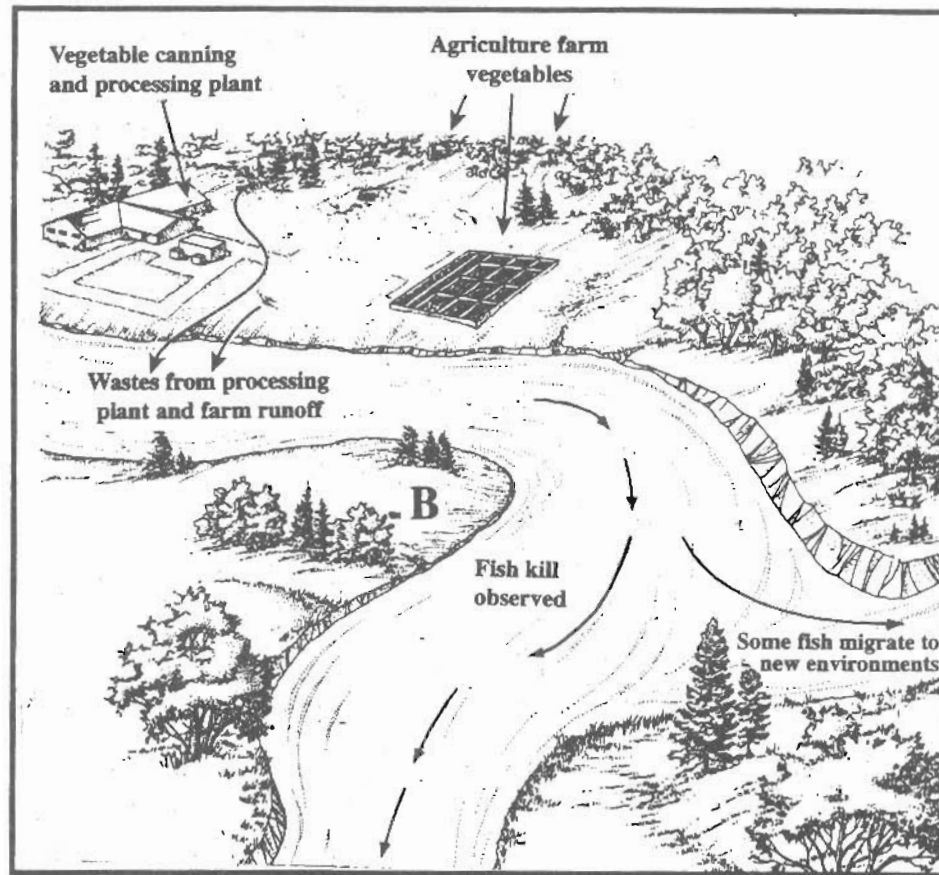


Figure 1. Site map

One of the first industries to be established was a vegetable farm and vegetable processing plant. The developer did not undertake an Environmental Impact Assessment before the farm and processing plant were established. Figure 2 shows the pyramid of energy and the biomass of organisms at the different trophic levels in the aquatic system (river).

Tertiary consumers	10	Humans	Biomass 500
Secondary Consumers	100	Fish	1000
Primary Consumers	1000	Zooplankton	1500
Producers Phytoplankton	10 000	Phytoplankton	3000

Figure 2. Pyramid of energy and biomass of organisms

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- (a) Using the information provided in Figure 2, explain how productivity in the ecosystem is achieved. [6 marks]
- (b) (i) Using the data from Figure 2, draw a pyramid of biomass to represent the trophic levels in the river ecosystem. [6 marks]
- (ii) Describe the procedure for determining the biomass of the fish indicated in Figure 2. [9 marks]
- (iii) Identify FOUR factors that influence 'Gross Primary Productivity'. [4 marks]
- (iv) Account for the trend shown in the pyramid of biomass drawn in (b) (i) above. [5 marks]

Total 30 marks

2. (a) Describe the sequence of events that may have resulted in the fish kills observed by residents of Community B. [8 marks]
- (b) (i) Identify TWO impacts resulting from the events described in 2 (a) above which may be of concern to residents of Community B. [2 marks]
- (ii) For EACH impact identified in (b) (i) above, outline ONE possible cause. [10 marks]
- (c) Outline the impact on the ecological balance of the aquatic community as a result of the population of fish kills and subsequent decline in fish population. [6 marks]
- (d) Figure 2 shows that only a small percentage of the energy absorbed by the phytoplankton is incorporated into the tissues of the secondary consumer. Provide TWO reasons for this. [4 marks]

Total 30 marks

3. Residents of Community B and the developer were told by an environmental officer that the problem of fish kills could have been avoided if an Environmental Impact Assessment was conducted before the development took place. The developer was also required to monitor the fish population and other mobile organisms in the river for a given period of time.
- (a) Suggest FOUR ways in which an Environmental Impact Assessment could have helped to mitigate the impacts being experienced. [8 marks]
- (b) Outline the MOST appropriate sampling technique that could be used to estimate population size of mobile (moving) populations. [7 marks]
- (c) (i) Describe THREE measures that can be implemented to prevent a recurrence of the negative environmental impacts and to protect fish and other aquatic species. [6 marks]
- (ii) Justify EACH measure described in (c) (i) above. [9 marks]

Total 30 marks

END OF TEST

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